

Computational improvements for clone-censor-weighting applications

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Disclosures



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- Catherine Wiener has received payment from Target RWE for unrelated work



Generative Artificial Intelligence Disclosure

I used generative artificial intelligence (GitHub CoPilot) in this study to assist in improving existing code, vectorizing existing code, and diagnosing/debugging coding errors. I am responsible for any mistakes and errors made.

Transparency and Reproducibility Statement

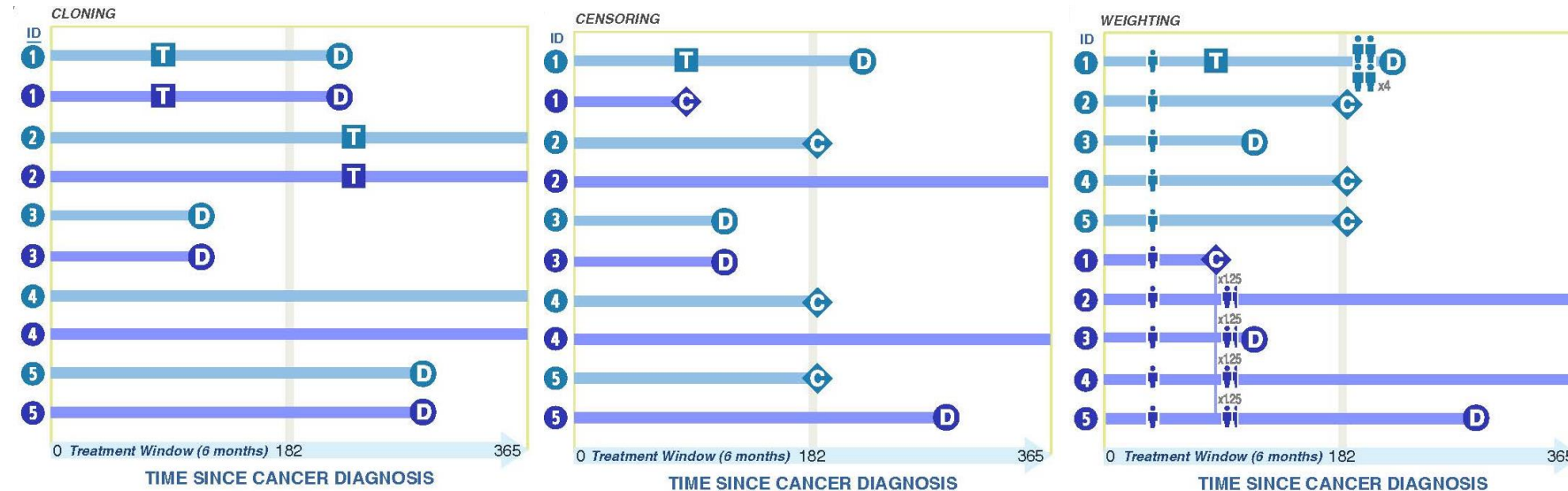
Analytical code with which to create the simulated cohorts, tables, and analysis results is available at:

https://github.com/jeannyww/CCW_EE

Clone-Censor-(Inverse Probability) Weight Method



- Estimates per-protocol effects for treatment strategies indistinguishable at eligibility
- Clone to assigned strategy, censor at non-adherence, IP weighted per-protocol analysis



Cain LE, Robins JM, Lanoy E, Logan R, Costagliola D, Hernán MA. When to Start Treatment? A Systematic Approach to the Comparison of Dynamic Regimes Using Observational Data. The International Journal of Biostatistics. 2010;6(2)doi:doi:10.2202/1557-4679.1212

Fu EL, Harhay MO, Schneeweiss S, Desai RJ, Hernán MA. Starting right: aligning eligibility and treatment assignment at time zero when emulating a target trial. BMJ. 2026;392:e084909. doi:10.1136/bmj-2025-084909

Figures: Dr. Jennifer Lund, UNC Lineberger CHAI Core, published in Gaber CE, Ghazarian AA, Strassle PD, et al. De-Mystifying the Clone-Censor-Weight Method for Causal Research Using Observational Data: A Primer for Cancer Researchers. Cancer Medicine. 2024;13(23):e70461. doi:https://doi.org/10.1002/cam4.70461

Statistical Inference for CCW method



- Maximum likelihood estimation approach does not account for:
 - Several sequential estimation steps
 - Multiple discrete time intervals and treatment strategies
 - Inverse probability weights estimated via nuisance models
- Non-parametric bootstrap for valid statistical inference
 - Computationally intensive for large datasets

Cain LE, Robins JM, Lanoy E, Logan R, Costagliola D, Hernán MA. When to Start Treatment? A Systematic Approach to the Comparison of Dynamic Regimes Using Observational Data. *The International Journal of Biostatistics*. 2010;6(2)doi:doi:10.2202/1557-4679.1212

Fu EL, Harhay MO, Schneeweiss S, Desai RJ, Hernán MA. Starting right: aligning eligibility and treatment assignment at time zero when emulating a target 5 trial. *BMJ*. 2026;392:e084909. doi:10.1136/bmj-2025-084909

Estimating Equations (EEs) as valid alternative for inference



- Define $\hat{\theta}$ ('M-estimator') as the solution to the EE:

$$\sum_{i=1}^n \psi(O_i; \hat{\theta}) = 0$$

- Large-sample variance of $\hat{\theta}$ via empirical sandwich variance estimator

Example: Sample mean ($\hat{\mu}$) for $i = 1 \dots n$ units of Y_i



$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^n Y_i$$

Rearrange terms to obtain the following EE:

$$\sum_{i=1}^n (Y_i - \hat{\mu}) = 0$$

EE: Estimating Equation (also referred to as M-Estimation)

Ross RK, Zivich PN, Stringer JSA, Cole SR. M-estimation for common epidemiological measures: introduction and applied examples. Int J Epidemiol. Feb 14 2024

Stefanski LA, Boos DD. The Calculus of M-Estimation. The American Statistician. 2002;56(1):29-38.

Estimating equations simultaneously account for multiple sources of variance



- Stacking estimating functions carries the variance between parameters

$$\boldsymbol{\psi}(O_i; \boldsymbol{\theta}) = \begin{pmatrix} \psi_1(O_i; \alpha) \\ \psi_2(O_i; \beta_1) \\ \psi_3(O_i; \beta_0) \\ \psi_4(O_i; \mu_1) \\ \psi_5(O_i; \mu_0) \\ \psi_6(O_i; \tau) \end{pmatrix}$$

Stefanski LA, Boos DD. The Calculus of M-Estimation. The American Statistician. 2002;56(1):29-38.

Ross RK, Zivich PN, Stringer JSA, Cole SR. M-estimation for common epidemiological measures: introduction and applied examples.

Int J Epidemiol. Feb 14 2024

A stack of EEs for CCW would parallel the estimation steps



$$\psi(O_i; \theta) = \begin{pmatrix} \psi_1(O_i; \alpha) & 1: \text{PLR (pooled logistic regression) for IP weights} \\ \psi_2(O_i; \beta_1) & 2: \text{Weighted PLR for Strategy 1 outcomes} \\ \psi_3(O_i; \beta_0) & 3: \text{Weighted PLR for Strategy 0 outcomes} \\ \psi_4(O_i; \mu_1) & 4: \text{Marginal risk under strategy 1} \\ \psi_5(O_i; \mu_0) & 5: \text{Marginal risk under strategy 0} \\ \psi_6(O_i; \tau) & 6: \text{Risk difference} \end{pmatrix}$$

- Estimate IP weights (unexpanded data)
- Clone to assigned treatment strategy
- Censor when deviate from assigned strategy
- Estimates of interest obtained from IP-weighted marginal structural model (on expanded data)

EE: Estimating Equations (also referred to as M-Estimation); PLR: pooled logistic regression; IP: inverse probability

Stefanski LA, Boos DD. The Calculus of M-Estimation. The American Statistician. 2002;56(1):29-38.

Zivich PN, Cole SR, Shook-Sa BE, DeMonte JB, Edwards JK. Estimating equations for survival analysis with pooled logistic regression. 2025/04/17 2025;

doi:<https://doi.org/10.48550/arXiv.2504.13291>

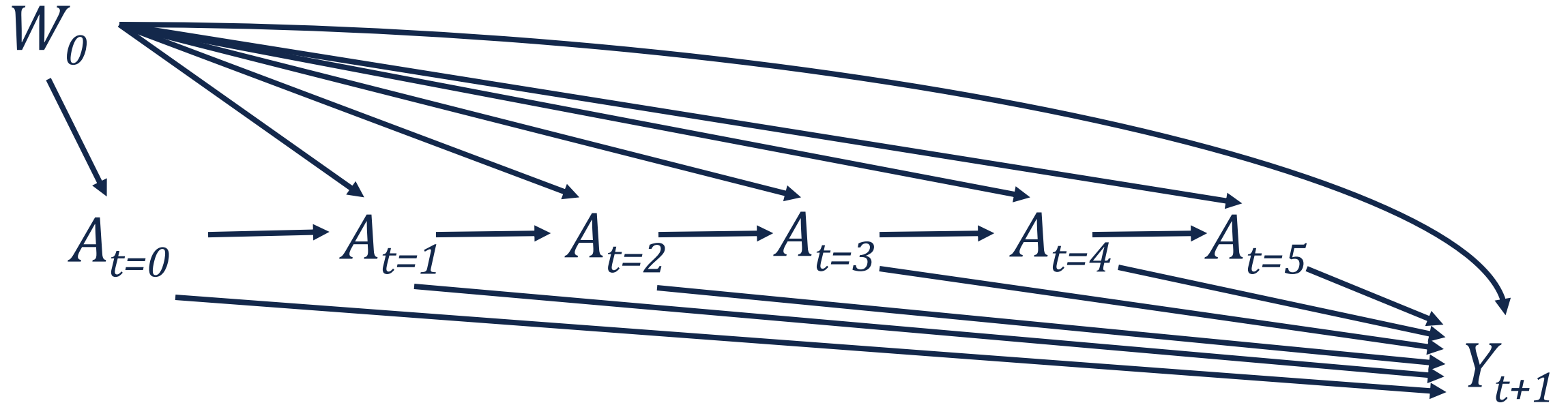


**To demonstrate variance estimation via
estimating equations for the CCW method**

Simulated Cohort Data Generating Process



- $n = 3,000$ patients: $i = 1, \dots, n$, of binary $O_i = (W_0, A_{i,t}, Y_{i,t+1})$ for $t \in [0, \dots, 5]$



- For grace period $m = 2$, simulate true effect and observed data
- Compare the effect of two treatment strategies on mortality, Y_{t+1}
 - **Strategy 0:** $A_t = 0$ for $t \leq m$
 - **Strategy 1:** $A_t = \begin{cases} \text{natural course:} & t < m \\ 1: & t = m \end{cases}$

Statistical Analysis



1. Evaluate EE performance for risk function and risk difference via Monte Carlo simulation
2. On a single dataset of $n=3,000$:
 - Execute clone-censor-(IP) weight analysis
 - Estimate variance using EE versus non-parametric bootstrap ($10^3, 10^5$ bootstraps)
3. Compare standard errors and elapsed computation time

Python 3.12, with Delicatessen 4.2

Red Hat Linux Server, single node, AMD EPYC 9354 processor, 128 GB Memory

Code: github.com/jeannyww/CCW_EE

EE: Estimating Equations (also referred to as M-Estimation)

Zivich PN, Klose M, Cole SR, Edwards JK, & Shook-Sa BE. (2022). *Delicatessen: M-Estimation in Python*. arXiv preprint arXiv:2203.11300.

See also: Barrett M (2026). *deli: M-Estimation and Empirical Sandwich Variance Estimation*. R package version 0.1.0, <https://github.com/r-causal/deli>.

EE Performance Metrics



Monte Carlo
simulation over
10,000 simulated
cohorts
(n=3,000 each)

Target Estimand	Event time	True Value	Bias	ESE	SER	95% CI Coverage
Marginal Risk under Strategy 0	1	0.10	0.001	0.006	0.98	0.947
	2	0.11	0.001	0.006	0.99	0.948
	3	0.12	0.001	0.007	0.99	0.947
	4	0.13	0.001	0.007	0.99	0.946
	5	0.15	0.001	0.007	0.99	0.947
	6	0.16	0.001	0.008	0.99	0.948
Marginal Risk under Strategy 1	1	0.12	0.000	0.006	0.99	0.948
	2	0.12	0.000	0.006	1.00	0.95
	3	0.14	0.001	0.009	0.99	0.947
	4	0.15	0.001	0.01	0.99	0.944
	5	0.16	0.001	0.011	0.99	0.946
	6	0.17	0.000	0.012	0.99	0.941
Marginal Risk Difference	1	0.011	-0.001	0.003	0.96	0.926
	2	0.012	-0.001	0.003	0.97	0.934
	3	0.017	0.000	0.007	0.98	0.926
	4	0.015	0.000	0.009	0.99	0.937
	5	0.013	0.000	0.01	0.99	0.939
	6	0.011	-0.001	0.012	0.99	0.943

EE Performance Metrics



Monte Carlo
simulation over
10,000 simulated
cohorts
(n=10,000 each)

Target Estimand	Event time	True Value	Bias	ESE	SER	95% CI Coverage
Marginal Risk under Strategy 0	1	0.10	0.001	0.003	0.96	0.94
	2	0.11	0.001	0.004	0.97	0.944
	3	0.12	0.001	0.004	0.98	0.945
	4	0.13	0.001	0.004	0.98	0.946
	5	0.15	0.001	0.004	0.98	0.946
	6	0.16	0.001	0.004	0.98	0.948
Marginal Risk under Strategy 1	1	0.12	0.000	0.003	0.99	0.946
	2	0.12	0.000	0.003	0.99	0.946
	3	0.14	0.001	0.005	0.99	0.948
	4	0.15	0.001	0.005	0.99	0.947
	5	0.16	0.001	0.006	0.99	0.947
	6	0.17	0.001	0.007	0.99	0.951
Marginal Risk Difference	1	0.011	-0.001	0.002	0.88	0.905
	2	0.012	-0.001	0.002	0.92	0.925
	3	0.017	0.000	0.004	1.00	0.944
	4	0.015	0.000	0.005	1.00	0.94
	5	0.013	0.000	0.006	1.00	0.946
	6	0.011	0.000	0.006	1.00	0.947

EE: estimating equations (also referred to as M-Estimation); ESE: empirical standard error; SER: standard error ratio; CI: confidence interval 14

Red Hat Linux Server OS, single node, AMD EPYC 9354 processor, 128 GB Memory

Estimating Equations versus Bootstrap Single dataset (n=3000) comparison

Target Estimand	Event time	Point Estimate	Std Err: Estimating Equations 0.76 seconds	Std Err: Bootstrap 1,000 iterations 211 seconds	
Marginal Risk under Strategy 0	1	0.101	0.00593	0.00594	Runtime on: • Red Hat Linux OS • AMD EPYC 9354 processor • 128 GB Memory • Single node • Single core
	2	0.111	0.00621	0.00631	
	3	0.123	0.00660	0.00672	
	4	0.138	0.00704	0.00717	
	5	0.154	0.00746	0.00750	
	6	0.166	0.00775	0.00782	
Marginal Risk under Strategy 1	1	0.111	0.00574	0.00578	
	2	0.121	0.00596	0.00610	
	3	0.142	0.00944	0.00940	
	4	0.150	0.01022	0.01011	
	5	0.163	0.01158	0.01160	
	6	0.174	0.01246	0.01241	
Marginal Risk Difference	1	0.010	0.00274	0.00273	
	2	0.011	0.00293	0.00290	
	3	0.020	0.00835	0.00815	
	4	0.012	0.00962	0.00927	
	5	0.009	0.01141	0.01115	
	6	0.008	0.01254	0.01231	

Estimating Equations versus Bootstrap Single dataset (n=3000) comparison

Target Estimand	Event time	Point Estimate	Std Err: Estimating Equations 0.76 seconds	Std Err: Bootstrap 100,000 iterations ~6 hours	Runtime on:
Marginal Risk under Strategy 0	1	0.101	0.00593	0.00593	
	2	0.111	0.00621	0.00621	• Red Hat Linux OS • AMD EPYC 9354 processor • 128 GB Memory • Single node • Single core
	3	0.123	0.00660	0.00662	
	4	0.138	0.00704	0.00705	
	5	0.154	0.00746	0.00748	
	6	0.166	0.00775	0.00778	
Marginal Risk under Strategy 1	1	0.111	0.00574	0.00573	
	2	0.121	0.00596	0.00595	
	3	0.142	0.00944	0.00946	
	4	0.150	0.01022	0.01024	
	5	0.163	0.01158	0.01161	
	6	0.174	0.01246	0.01250	
Marginal Risk Difference	1	0.010	0.00274	0.00275	
	2	0.011	0.00293	0.00293	
	3	0.020	0.00835	0.00837	
	4	0.012	0.00962	0.00963	
	5	0.009	0.01141	0.01141	
	6	0.008	0.01254	0.01255	

Is this EE a valid alternative?



- Explored with minimum example simulation
- EEs rely on large sample theory
 - Potential trade-off for reduced computation burden
- Culprits of non-nominal coverage:
 - Simulating correct counterfactual pseudo-population
 - Bias due to model specification

EE: estimating equations (also referred to as M-Estimation)

Stefanski LA, Boos DD. The Calculus of M-Estimation. The American Statistician. 2002;56(1):29-38.

Next steps: Refine EE application scope



- Decipher sources of non-nominal 95% CI coverage
- SAS implementation
- 'IPCW' implementation
- Treatment strategy complexity and modeling flexibility
- Empirical application

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SW-2G - Fake World Data, Real World Problems—Simulations for the pharmacoepidemiologist

Tonight! 5:15 PM - 6:45 PM CEST, Space 4, Level 0

SW-3D - How I Learned to Stop Worrying and Start Including People in Studies Multiple Times Tuesday, September 1, 2026, 1:00 PM - 2:30 PM CEST

C-350 - Standard Error Estimation for Transported Effects in Sequential Nested Trial Emulations Using M-Estimation

Wednesday, September 2, 2026, 12:00 PM - 1:30 PM CEST

6A-06 - Data Fusion and M-Estimation for Pharmacoepidemiology: a demonstration and simulation study: Wednesday, September 2, 2026 2:45 PM - 3:00 PM CEST



Thank you!

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https://github.com/jeannyww/CCW_EE





Empirical Sandwich Variance

- The estimating function $\psi(O_i, \theta)$ that solves $E[\psi(O_i, \theta)] = 0$
- An estimator $\hat{\theta}$ for θ is the solution to $n^{-1} \sum_{i=1}^n \psi(O_i; \hat{\theta}) = 0$
- Asymptotic variance of $\hat{\theta}$ can be consistently estimated* by:

$$V_{\hat{\theta}} = B(\hat{\theta})^{-1} F(\hat{\theta}) \left(B(\hat{\theta})^{-1} \right)^T$$

$$B(\hat{\theta}) = \frac{1}{n} \sum_{i=1}^n \left[\frac{\partial \psi(O_i; \theta)}{\partial \theta} \right]$$

$$F(\hat{\theta}) = \frac{1}{n} \sum_{i=1}^n \psi(O_i; \hat{\theta}) \psi(O_i; \hat{\theta})^T$$

*Under suitable regularity conditions
Stefanski LA, Boos DD. The Calculus of M-Estimation.
The American Statistician. 2002;56(1):29-38.

An Example: An EE for the Sample Mean



- We want to estimate the sample mean ($\hat{\mu}$) of $i = 1 \dots n$ units of Y_i

We could use:

$$n^{-1} \sum_{i=1}^n Y_i = \hat{\mu}$$

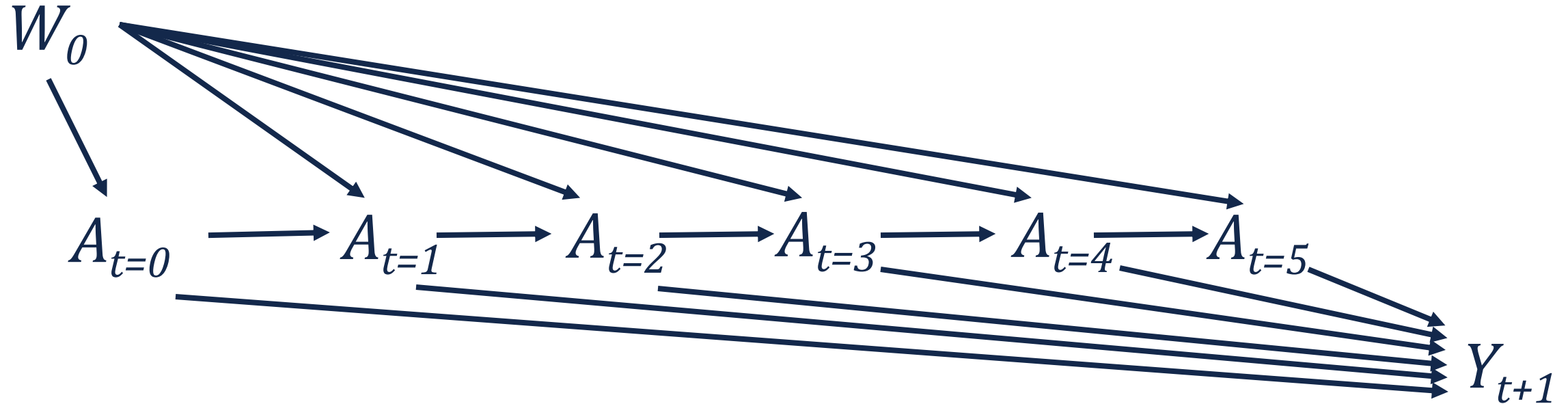
Or rearrange terms:

$$n^{-1} \sum_{i=1}^n Y_i - \hat{\mu} = \hat{\mu} - \hat{\mu}$$

For an EE for the sample mean:

$$n^{-1} \sum_{i=1}^n (Y_i - \hat{\mu}) = 0$$

Simulated Cohort (Detailed Data Generation Process)



Units $i = 1, \dots, n$, comprised of $O_i = (A_i, Y_i, W_0)$ for $t \in [0, \dots, 5]$

$A_0 \sim \text{Bernoulli}(p = \text{expit}(-2 + 0.5W_0))$

$A_{t>0} \sim \text{Bernoulli}(p = \text{expit}(-2 - 0.8A_{t-1} - 0.5W_0 + 0.1t))$

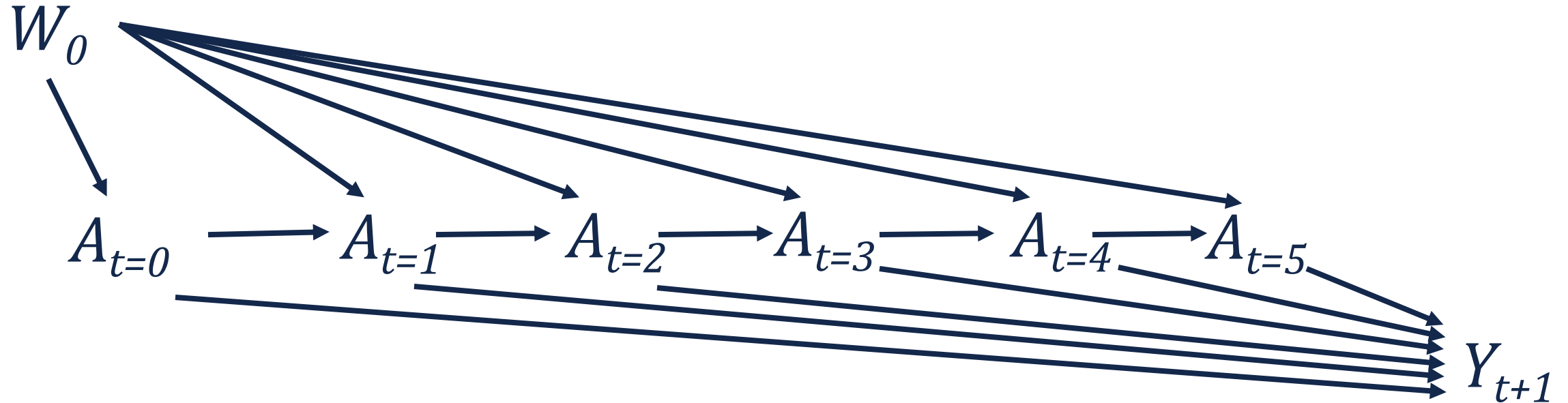
$Y_{t=1} \sim \text{Bernoulli}(p = \text{expit}(-2 + 0.7A_{t-1} - 0.6W_0))$

$Y_{t>1} \sim \text{Bernoulli}(p = \text{expit}(-2 + 0.6A_{t-1} + 0.7W_0 + 0.1t))$

Simulated Cohort



- 3,000 patients $i = 1, \dots, n$, comprised of binary $O_i = (W_0, A_{i,t}, Y_{i,t+1})$ for $t \in [0, \dots, 5]$



- Compare the effect of treatment strategy 1 vs 0 on mortality, Y_{t+1}
 - Strategy 0:** $A_t = 0$ for $t \leq m$
 - Strategy 1:** $A_t = \begin{cases} \text{natural course, } t < m \\ 1, t = m \end{cases}$

Identification Assumptions



Identification Assumption	
Causal consistency	For treatment strategy $\bar{A} = \bar{a}$, $Y_{t+1}^{\bar{a}} = Y_{t+1}$ if $C_t = 0$
Conditional exchangeability for treatment strategy	$Y_{t+1}^{\bar{a}} \perp A_t W_0, \bar{A}_{t-1}, Y_t = 0$
Conditional exchangeability for treatment strategy implies conditional exchangeability for artificial censoring	$Y_{t+1}^{\bar{a}} \perp C_t^{\bar{a}} C_{t-1}^{\bar{a}} = 0, W_0, \bar{A}_{t-1}, Y_t = 0$
Positivity for treatment strategy	$1 > \Pr(A_t = 1 W_0, \bar{A}_{t-1}, Y_t = 0) > 0$ wp1
Positivity for treatment strategy implies positivity for artificial censoring	$1 > \Pr(C_t^{\bar{a}} = 0 C_{t-1}^{\bar{a}} = 0, W_0, \bar{A}_{t-1}, Y_t = 0) > 0$ wp1

A stack of EEs for CCW for the estimation steps



Estimators $\hat{\theta} = (\hat{\alpha}, \dots, \hat{\tau})$ are the solution to $\sum \psi(O_i; \theta) = 0$

$$\psi(O_i; \theta) = \begin{pmatrix} \psi_1(O_i; \alpha) \\ \psi_2(O_i; \beta_1) \\ \psi_3(O_i; \beta_0) \\ \psi_4(O_i; \mu_1) \\ \psi_5(O_i; \mu_0) \\ \psi_6(O_i; \tau) \end{pmatrix}$$

- 1: Pooled logistic regression (PLR) EE for estimating time-varying IP weights
- 2: Weighted PLR EE for the discrete hazards of outcome among strategy 1 clones
- 3: Weighted PLR EE for the discrete hazards of outcome among strategy 0 clones
4. EE for marginal risk of outcome among strategy 1 clones
5. EE for marginal risk of outcome among strategy 0 clones
6. EE for risk difference of outcome between clones of strategy 1 and 0

- Estimate time-varying IP weights (unexpanded data)
- Clone to assigned treatment strategy
- Censor when deviate from assigned strategy
- Estimates of interest obtained from IP weighted marginal structural model (on expanded data)

Estimating Equations – IPTW approach

Observations $i = 1, \dots, n$ with observed data $O_{i,k}$

$O_{i,k} = (A_{i,k}, \bar{A}_{i,k-1}, T_{i,k}, C_{i,k}^a, Y_{i,k}, W_{i,0})$

K = number of discrete time intervals

$k \in K = \{1, \dots, K\}$

$A_{i,k}$ = observed binary treatment action at time k

T_i = event time

C_i^a = artificial censoring time for nonadherence to treatment under strategy assigned

$T_i^{a*} = \min(T_i, C_i^a)$ under treatment strategy assigned

$Y_{i,k}$ = binary outcome at time k

$W_{i,0}$ = binary time-invariant covariate

$I(T_i > s_{k-1})$ = risk set

$I(T_i^{a*} > s_{k-1})$

= risk set under strategy-specific artificial censoring

$\mathbb{X}_{i,k} = [W_{i,0} \bar{A}_{i,k-1}]$

$\mathbb{S}_{i,k}$ = [time design matrix]

$\theta = (\alpha, \dots, \tau)$

α = parameters for time-varying treatment model

β_1 = parameters for discrete time hazard of event under strategy 1

β_0 = parameters for discrete time hazard of event under strategy 0

$\mu_{1,k}$ = marginal risk at time k for strategy 1

$\mu_{0,k}$ = marginal risk at time k for strategy 0

τ_k = marginal risk difference at time k

$\hat{A}_{i,k} = \text{expit}([\mathbb{X}_{i,k} \mathbb{S}_{i,k}] \alpha^T)$

$\hat{Y}_{i,k}^a = \text{expit}([\mathbb{S}_{i,k}] (\mathbf{B}_1)^T)$

$\hat{Y}_{i,k}^a = \text{expit}([\mathbb{S}_{i,k}] (\mathbf{B}_0)^T)$

$$\omega_{i,k}^a = \left\{ \prod_{k \in K} (\Pr(A_k = a_k | \bar{A}_{k-1} = \bar{a}_{k-1}, \bar{A}_{k-1} = 0, \bar{C}_{k-1} = 0, W = w)) \right\}^{-1}$$

$$\psi_{IPTW}(O_i; \theta) = \begin{pmatrix} \psi_1(O_i; \alpha) \\ \psi_2(O_i; \beta_1) \\ \psi_3(O_i; \beta_0) \\ \psi_4(O_i; \mu_1) \\ \psi_5(O_i; \mu_0) \\ \psi_6(O_i; \tau) \end{pmatrix} =$$

$$\begin{pmatrix} \sum_{k \in K} \{I(T_i > s_{k-1}) (A_{i,k} - \hat{A}_{i,k}) [\mathbb{X}_{i,k} \mathbb{S}_{i,k}]^T\} \\ \sum_{k \in K} \{I(T_i^{a=1*} > s_{k-1}) (Y_{i,k} - \hat{Y}_{i,k}^{a=1}) [\mathbb{S}_{i,k}]^T\} \omega_{i,k}^{a=1} \\ \sum_{k \in K} \{I(T_i^{a=0*} > s_{k-1}) (Y_{i,k} - \hat{Y}_{i,k}^{a=0}) [\mathbb{S}_{i,k}]^T\} \omega_{i,k}^{a=0} \\ \left\{ 1 - \prod_{k \in K} (1 - \hat{Y}_{i,k}^{a=1}) \right\} - \mu_{1,k} \\ \left\{ 1 - \prod_{k \in K} (1 - \hat{Y}_{i,k}^{a=0}) \right\} - \mu_{0,k} \\ (\mu_{1,k} - \mu_{0,k}) - \tau_k \end{pmatrix}$$

Ross RK, Zivich PN, Stringer JSA, Cole SR. M-estimation for common epidemiological measures:

introduction and applied examples. *Int J Epidemiol.* Feb 14 2024;53(2):doi:10.1093/ije/dyae030

Zivich PN, Cole SR, Shook-Sa BE, DeMonte JB, Edwards JK. Estimating equations for survival analysis with pooled logistic regression. 2025/04/17 2025;doi:https://doi.org/10.48550/arXiv.2504.13291

Estimating Equation versus Bootstrap Single dataset (n=3000) comparison

Target Estimand	Event time	Point Estimate	Std Err: EE 0.76 seconds	Std Err: Bootstrap 10 ⁵ iters 21,122 seconds	Std Err: Bootstrap 10 ⁴ iters 2,162 seconds	Std Err: Bootstrap 10 ³ iters 211 seconds
Marginal Risk under Strategy 0	1	0.101	0.006	0.006	0.006	0.006
	2	0.111	0.006	0.006	0.006	0.006
	3	0.123	0.007	0.007	0.007	0.007
	4	0.138	0.007	0.007	0.007	0.007
	5	0.154	0.007	0.007	0.007	0.008
	6	0.166	0.008	0.008	0.008	0.008
Marginal Risk under Strategy 1	1	0.111	0.006	0.006	0.006	0.006
	2	0.121	0.006	0.006	0.006	0.006
	3	0.142	0.009	0.009	0.009	0.009
	4	0.150	0.010	0.010	0.010	0.010
	5	0.163	0.012	0.012	0.012	0.012
	6	0.174	0.012	0.012	0.013	0.012
Marginal Risk Difference	1	0.010	0.003	0.003	0.003	0.003
	2	0.011	0.003	0.003	0.003	0.003
	3	0.020	0.008	0.008	0.008	0.008
	4	0.012	0.010	0.010	0.009	0.009
	5	0.009	0.011	0.011	0.011	0.011
	6	0.008	0.013	0.013	0.012	0.012

EE: estimating equations, Std Err: Standard Error, iters: Iterations